



Spatial Analysis in Business

How Geospatial Intelligence is Transforming Global Operations Strategy and Risk Management

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What if your company could have predicted the last major supply chain disruption before it happened?

The Geographic Information Systems (GIS) market is set to explode with a projected surge of growth from \$12.9 billion in 2024 to \$32.04 billion by 2033.¹ As Figure 1 demonstrates, this is a massive 148% expansion in less than a decade, driven by a compound annual growth rate (CAGR) of 10.64%.

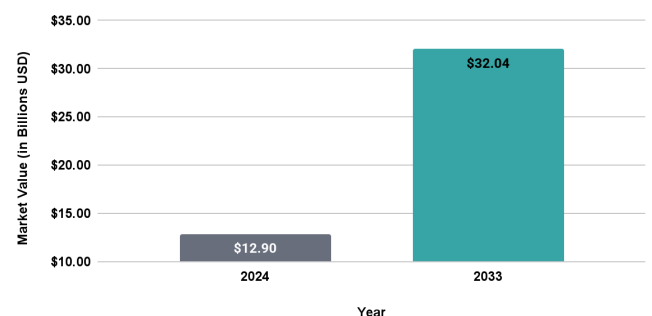
But what is truly fueling this tremendous expansion?

Navigating the modern supply chain proves to be more complex than ever with geopolitical tensions are higher than ever,

increased climate risks, and complex problems with infrastructure.

What happens to businesses when disaster strikes, halting supply chains in Southeast Asia, or when new trade tariffs unforeseen by traditional supply chain analysis methods disrupt distribution networks?

Figure 1: Geographic Information Systems Market Value (Billions USD) Source: M2PressWIRE, 2025



Now, more than ever, business leaders need to prepare for these complex spatial phenomena to ensure that their business has the tools to succeed.

What is Spatial Analysis and Why Does it Matter?

Geographic information systems (GIS) have long served as the robust backbone of many government agencies and environmental scientists. GIS is "a technology that is used to create, manage, analyze, and map all types of data."²

Business leaders and professionals are discovering the connection between this spatial data and its applicability to core business functions such as optimizing supply chains, analyzing risks, improving infrastructure, and ultimately generating greater financial returns.



Every business is connected to some type of spatial data.

Whether it is mapping site locations, customer foot traffic or sourcing products,

spatial data is a critical component of making informed decisions. Geographic Information Systems is the bridge that businesses can use to understand these spatial patterns or connections. A GIS is technology that creates, manages, analyzes, and maps data. Rather than looking at a spreadsheet filled with names or numbers and agonizing over columns and rows, GIS allows users to visualize these complex relationships through maps; revealing trends and valuable analysis not seen by traditional analysis.

Business professionals regularly tasked with making critical decisions need every edge they can get. Imagine you are a retailer looking to expand and evaluate potential locations. A traditional method of analysis might look at various demographics, competition, and foot traffic data. Utilizing spatial analysis techniques, these variables can be easily mapped revealing relationships between factors and offering insights to give a competitive edge—gaps in competition, areas with heavy foot traffic and wealthier demographics. Location-based analytics powered by geographic information systems sheds light on these valuable relationships.

Applications for GIS in business are more than just putting data on maps. Organizations already utilizing geographic analytics have seen growth in key performance indicators including better risk identification and mitigation, more effective resource allocation, improved logistics, and massive reductions in costs.

Location-based analytics powered by business-centered geographic information systems offers practical insights and powerful tools to business professionals tasked with making decisions regarding global operations strategy and risk management.

Real World Application

Those early to GIS implementation are already being rewarded with substantial returns across various industries. Nationwide, organizations at every scale have seen impressive growth and returns using spatial analysis techniques powered by GIS.

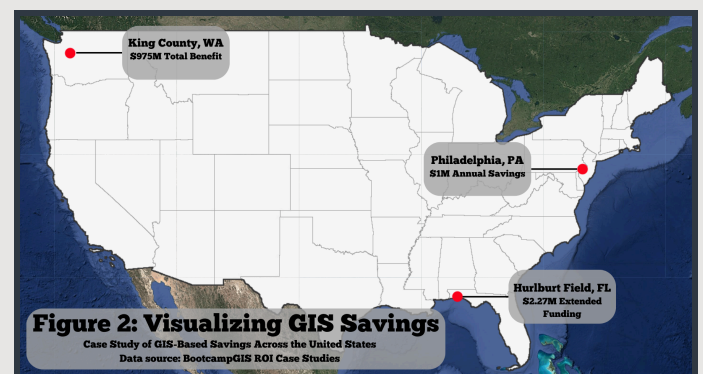
In King County, Washington an initial investment of \$200 million dollars has generated a massive \$975 million in net benefit³ over the past two decades. By utilizing GIS the municipality's investment has yielded greater efficiency in areas like emergency response planning, environmental conservation, and urban development. With spatial analytics proving useful in optimizing service routes, mitigating flood risks, and providing metrics to increase efficiency the county has optimized in ways previously unseen by traditional analysis and management.

In Philadelphia, the municipal government has utilized GIS to generate over \$1 million annually⁴ through optimizing LED traffic light placement and creating more effective maintenance schedules. The

rewards of this GIS-based project continue to compound each year.

Hurlburt Field Air Force Base located in Florida conducted facility audits using GIS and managed to identify \$2.27 million in extended funds.⁵ Their spatial analysis revealed inefficiencies within their infrastructure that had previously been overlooked by traditional inspections.

Value creation from GIS is even greater with spatial analytics being used to save the United States hundreds of billions yearly in the utilities and risk management sectors. With GIS being used to optimize grid infrastructure, and reduce outage response times, utility companies alone are benefitting massively from this spatial technology.



The widespread success of GIS implementations in states of Washington, Pennsylvania, and Florida demonstrate the universal applicability of this revolutionary technology.

Whether optimizing street lights in packed urban environments like Philadelphia or delivering returns from spatial analysis in

Florida, GIS stands out for its ability to generate measurable financial returns.

Supply Chain Risk Management

Modern supply chains are facing unprecedented volatility and challenges. With geopolitical tensions, climate disasters and infrastructure vulnerabilities ready to wreak havoc without warning. Traditional supply chain analysis and risk management techniques of comparing supplier costs and auditing for quality miss out on these make-or-break spatial phenomena completely.



Location-based analytics has the power to transform supply chain analysis shining a light on geographic trends or risks that remain hidden when examined by traditional methods. Companies utilizing GIS to map supplier networks tend to exhibit vulnerabilities with components sourced from typhoon-prone regions or suppliers clustered in areas of immense geopolitical tension. Without GIS these vulnerabilities can remain unseen and create massive problems for companies in

the face of disaster. Research by Goldberg and colleagues confirms this:

"Spatial analysis yields a powerful supply base risk mitigation strategy, empowering companies with knowledge of geographic factors to reduce risk exposure and improve overall supply chain resilience."

The power of spatial analysis in sourcing decisions is demonstrated brilliantly in research as Goldberg and company note spatial analysis yields "a powerful supply base risk mitigation strategy."⁶ Empowering companies with knowledge of geographic factors, business professionals can complete more in-depth analyses reducing risk exposure and improving overall supply chain resilience.

Outside of sourcing decisions, companies are using GIS for live risk monitoring of threats in their supply networks. Disasters such as hurricanes, wildfires, and political instability are closely monitored by GIS allowing for companies to immediately respond and identify alternative routes, modify orders, and proactively respond to potential supply chain catastrophes.

Operational Applications

Spatial analytics have the power to drive operational success across a variety of business functions. Site selection decisions—knowing where to construct facilities, open retail stores, or where to

place critical infrastructure like distribution centers are high stakes decisions that can have long-term effects on a company. GIS turns these decisions from educated guesses to real data-backed strategies.

GIS in retail site planning is a perfect example of geospatial intelligence in use with complex analysis of demographics, competition, and foot traffic companies can visualize and make multi-criteria decisions. Knowledge emerging from this geographic analysis provides businesses with a competitive edge, knowing where market gaps occur, and identifying optimal locations for sites. Geomarketing research shows how GIS is enabling “specific targeting strategies and operational efficiency” that is missed by traditional market research.

NVIDIA, tech superpower, has pioneered AI-powered spatial analysis for industrial automation, using real-time mapping and digital twins to solve complex routing problems.⁸ Their massive distribution network made up of thousands of points is carefully examined by spatial algorithms that can help identify optimal routes, minimizing costs and improving overall business efficiency.

Gains from improvements in efficiency will continue to compound over time, creating greater profits and helping companies to keep up with increases in operational complexity. Organizations like NVIDIA that can master this powerful spatial technology could be setting themselves up for compounding gains while competitors

struggle with traditional analysis techniques.

Explosive Growth Trajectory

The market value of GIS is projected to grow from \$12.9 billion in 2024 to \$32.04 billion by 2033 based on a 10.64% compound annual growth rate (CAGR)⁹. This growth reflects the massive shift taking place in how companies are operating and making decisions.

Advancements in artificial intelligence and machine learning combined with the increased availability of cloud computing have helped create greater access to complex spatial analysis, enhancing GIS capabilities. Techniques like predictive modeling, pattern recognition, and risk analysis can now be performed on greater scales. Data streams tracking real-time events from remote sensing satellites, mobile devices, and other sensors are providing a wealth of spatial data ripe for analysis.

With these advancements in technology bridging the gap for spatial analysis to enter into new industries. GIS is being utilized in financial services for credit assessment and fraud detection. Healthcare organizations are using location-based analysis to find optimal locations for their facilities and to map disease patterns. Telecommunications companies are continuing to improve infrastructure through spatial modeling and optimizing cell tower placement. GIS continues to prove itself as a viable

technology across diverse sectors with new use cases opening up the market and pushing technological boundaries.

Creating a Geospatial Advantage

Improvements in technical capabilities and technological infrastructure have elevated GIS to business criticality with businesses already using spatial analysis to generate measurable financial returns, create more resilient supply chains, and make more informed strategic decisions than those relying on traditional methods.

The market's explosive trajectory reflects this fundamental shift in global operations strategy and risk management. The climate of geopolitical tension and risk of disasters continues to create more complex decisions for businesses, the need for geospatial intelligence in business intensifies.¹⁰ Knowledge gained from GIS-based spatial analysis to analyze complex spatial relationships, and other spatial phenomena to optimize location-based decisions is no longer an option but the standard for businesses looking to succeed in this technological landscape.

Business professionals navigating modern supply chains, evaluating expansion opportunities, or optimizing operational efficiencies must ask themselves how they can quickly adopt and develop these spatial capabilities to better implement them into their decision making process.

Companies that can master geospatial intelligence will possess competitive advantages that will continue to compound, creating better decisions regarding locations, supply chains, risks, and opportunities. Companies that delay or deny this technology risk finding themselves at a great disadvantage in an economy where understanding the “where” has become just as important as understanding the “what.”

The geospatial revolution in business has already started.

The question that business leaders must consider: How quickly can we integrate geospatial intelligence into our own strategic planning? Those who are able to act quickly will build compounding advantages, while those who hesitate risk becoming irrelevant in an economy of understanding the ‘where’ is just as important as understanding ‘what.’

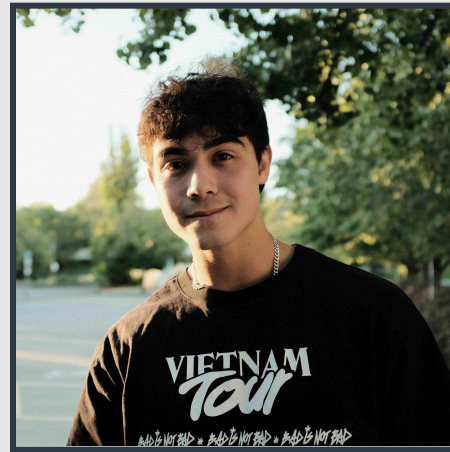
Endnotes

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About the Author



Author Devin Tran is a Geospatial Intelligence major at Brigham Young University. In addition to his studies in the field of Geography, Devin is also working towards completing a minor in Computer Science.

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